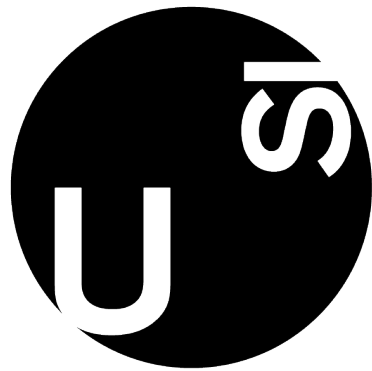




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# **Project 2**

## Data Analytics

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# 1 Introduction

The objective of this project is to develop a music recommendation system using Spotify data to personalize the listener experience. We performed an Exploratory Data Analysis to identify key drivers of musical preference. We pre-processed the features by handling missing values and scaling data to train two distinct recommendation models. Finally, we evaluated and compared their performance for delivering music suggestions.

## 2 Exploratory Data Analysis and Data Pre-processing

The dataset consists of five files, namely: *data.csv*, *data\_by\_artist.csv*, *data\_w\_genres.csv*, *data\_by\_year.csv*, *data\_by\_genres.csv*. After an inspection we noticed that the most important is the first one (*data.csv*) because every single row in this file represents one individual song, and it contains over 170'000 rows, including specific details the other files don't. The remaining files offer pre-aggregated insights: *data\_by\_artist.csv* and *data\_w\_genres.csv* groups them by artist, and averages out their audio features, the latter adding also genres, while *data\_by\_year.csv* averages the audio features for every song released in a given year, so it has just 100 entries. Finally, *data\_by\_genres.csv*, similar to the year and artist files, groups the data by specific musical genres, giving us the average audio profile of roughly 3,000 distinct genres.

So, for this exploratory part we will focus on the first file the most because it captures the full variance of the music library. As we can see in Figure 1 aggregated datasets (by artist, year, or genre) tend to collapse the diversity of music into a narrow mean. For example, while the average danceability in 1980 might be 0.56, *data.csv* shows that individual tracks that year ranged from 0.07 to 0.95. To build a recommendation system that can distinguish for example between a high-energy workout song and a low-energy study track by the same artist, the granular track-level data in *data.csv* is indispensable.

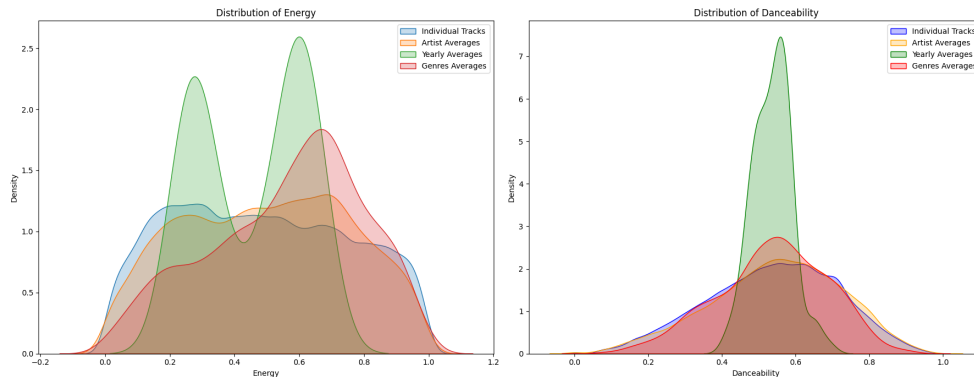


Figure 1: Granularity Inspection

Moving forward to *data.csv*, we controlled that it has no missing values, and then in Figure 2 we plot the heatmap. We have 19 columns, and we can organize them in subgroups: **Track Metadata** (id, name, artists, release\_date, year, duration\_ms, explicit); **Mood Vibes** (popularity, valence, energy, danceability, acousticness); **Technical Features** (tempo, loudness, mode, key, instrumentalness, speechiness, liveness);

Visual inspection of the matrix in Figure 2 confirms the distinct cluster of warmer colors in the upper-left corner, revealing notable intercorrelations between the Mood Vibes group and related technical elements like *loudness*. The single strongest positive relationship is found between the track metadata feature *year* and *popularity* (0.86), indicating newer additions to the dataset are generally more highly streamed. Strong expected ties are also evident between *energy* and *loudness* (0.78), and *valence* and *danceability* (0.56). Conversely, the starkest negative correlation (-0.75) occurs between *acousticness* and *energy* (intuitive inverse relationship between acoustic sound and perceived sonic intensity). Feature analysis reveals that *key* and *mode* exhibit the weakest correlations overall, showing essentially no linear connection to other features, while *acousticness* also demonstrates a notable negative trend against *year* (-0.61), suggesting music in this dataset has grown less acoustic over time.

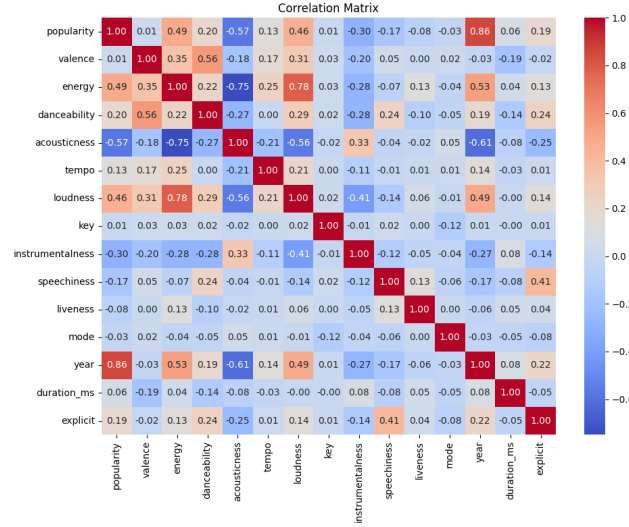


Figure 2: Correlation Matrix

## 2.1 Track Metadata

This category is useful for identifying a certain song but it does not convey a lot of information: `id` is the unique Spotify identifier for the track. `name` is the title of the track. `artists` are the name(s) of the artist(s) who performed the track. Then we have the `release_date` with the year (we think extracted for easier time-series analysis). We then have the `duration_ms` in milliseconds. Finally we have the variable *explicit* that is a binary flag indicating whether the track contains explicit content (1) or not (0). In this matter the most useful information about this group are the duration and the date of release that we will analyze in the next in the Data Cleaning section and in Other Datasets section respectively. Some of these features are included in Figure 4 correlated to others.

## 2.2 Mood Vibes

In this group the variables are calculated algorithmically to describe the subjective "feel" of the track. They are all normalized on a scale from 0.0 to 1.0. The popularity is a score from 0 to 100 calculated by the platform and based on the total number of recent streams. The valence stands for the musical "positiveness" of the track: high valence (closer to 1.0) sounds happy, while low valence (closer to 0.0) sounds sad. Then, as we saw in Figure 1 we have the energy that measures the intensity. Fast, loud, and noisy tracks (e.g., heavy metal) score near 1.0, while quiet, slow tracks (e.g., acoustic ballads) score near 0.0. And the danceability, meaning how suitable a track is for dancing, derived from a combination of technical features. Then we have acousticness that says whether the track is purely acoustic (1.0) or uses electronic instruments and synthesizers (closer to 0.0).

Valence and energy exhibit broad, relatively flat distributions, indicating that the dataset captures a highly balanced spectrum of emotional tones, from melancholy and mellow to euphoric and intense. Danceability follows a classic normal distribution, reflecting the natural tendency of most music to maintain a moderate, accessible groove rather than pushing rhythmic extremes. In stark contrast, acousticness is distinctly bimodal, revealing that tracks typically commit entirely to either electronic production or purely acoustic arrangements, largely abandoning middle-ground hybrid styles. Finally, the popularity metric exposes a severe disparity in music consumption; a massive volume of tracks remain completely undiscovered or forgotten at zero, while the surviving music follows a typical curve of moderate to high mainstream engagement.

## 2.3 Technical Features

Here instead we have variables describing the literal sound waves, structural elements, and recording environment of the track. In fact we have the tempo (pace of a track, Beats Per Minute), loudness (average loudness

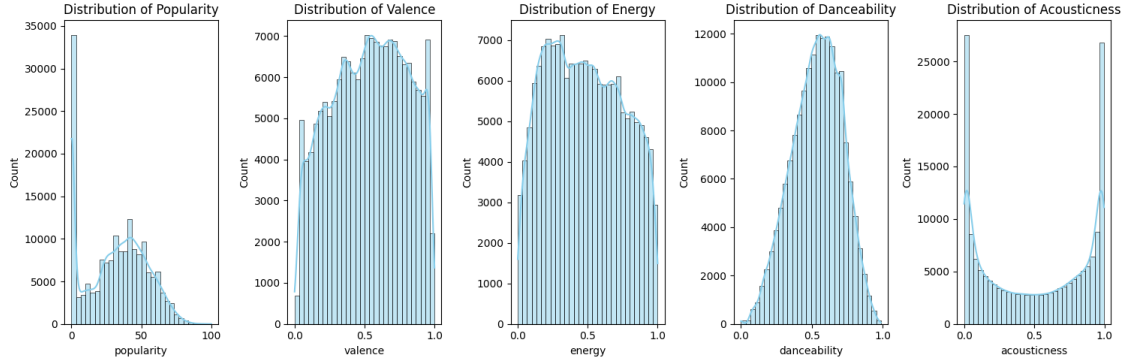


Figure 3: Mood Features Distributions

in decibels), the mode that indicates the modality (major or minor) of the track: major keys (1) usually sound brighter, while minor keys (0) sound darker. Then we have the key that estimates overall musical key of the track (e.g., C, C#, D) mapped to integers using standard Pitch Class notation. In addition we have instrumentality (score closer to 1.0 indicates an instrumental track without any singing). On the other hand the speechiness detects the presence of exclusively spoken words. We can later use high scores (above 0.6) to usually identify podcasts, stand-up comedy, or audiobooks rather than music. Finally the liveness that detects the presence of an audience during the recording (above 0.8 provides high confidence that the track was recorded at a live rather than in a studio).

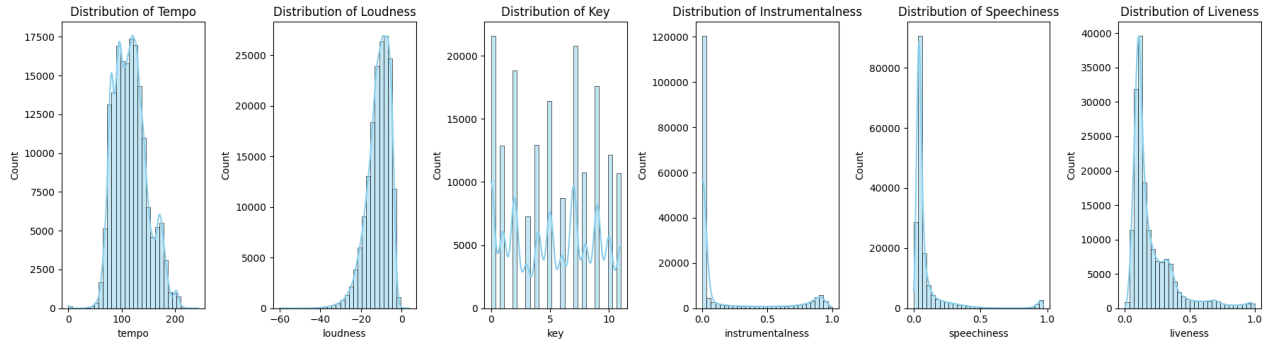


Figure 4: Technical Features Distributions

The heavy skew across instrumentality, speechiness, and liveness firmly establishes that the vast majority of this dataset consists of traditional, vocal-driven studio recordings rather than live concert tapes, spoken-word tracks, or pure instrumentals. Loudness is heavily pushed toward higher decibel levels, illustrating modern audio mastering trends where tracks are engineered to be as loud as possible to catch the listener’s attention. Meanwhile, tempo centers around the 100 to 130 BPM range, representing the foundational rhythmic of most pop and rock music. Finally, the distribution of musical keys reveals a compositional bias, proving that artists favor historically resonant and popular keys, like C Major (key 0) and G Major (key 7), over more complex tonal centers.

## 2.4 Other Datasets

Using *data\_by\_year.csv*, we can visualize how the “sound” of music has fundamentally shifted over the last century. For example acousticness dominated the early 20th century but took a massive nosedive as electric instruments and synthesizers were introduced. Energy and Danceability have steadily climbed, reflecting the rise of modern pop, rock, and electronic dance music. Parallel to these technical shifts, the second visualization compares the total volume of songs in the dataset against the prevalence of explicit content. While the number of recorded tracks stabilized at a high volume after the 1950s, the ratio of explicit tracks remained negligible for decades before beginning a sharp, consistent climb in the early 1990s, reaching its peak by 2020.

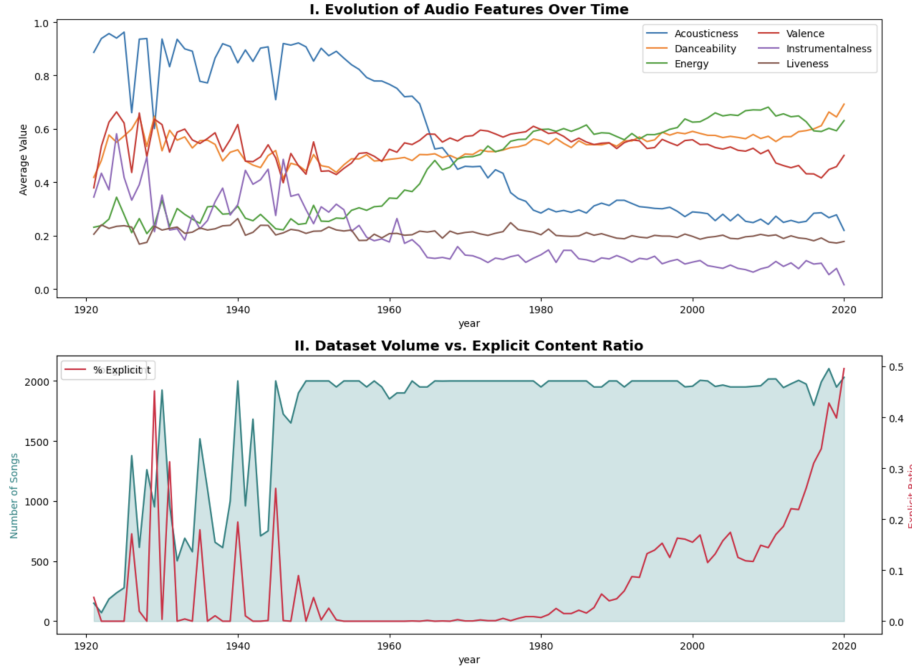


Figure 5: Features Evolution over time

In addition, from this dataset and from *data\_by\_genres.csv* we computed the Decade Most Popular Tracks and the Most Popular Genres that can be seen in the *python notebook*.

## 2.5 Outliers and Data Cleaning

Looking at outliers, we can see in the *notebook* that the variable with the higher number of outliers are: duration\_ms, tempo, loudness, instrumentalness, speechiness, liveness. To ensure the recommendation system operates on musically high-quality data, we enrich the *data.csv* with genre information using a primary-artist matching strategy, which provides the model with essential categorical context. To eliminate non-musical artifacts such as podcasts, white noise, and extreme technical anomalies, we applied logical thresholds to features like speechiness, duration, and loudness based on industry standards. Furthermore, by employing the IQR method and Min-Max scaling, we mitigated the influence of extreme outliers and brought disparate variables—such as tempo and danceability—into a uniform scale. In addition we remove duplicates (leaving the most popular version) and the zero value corruptions.



Figure 6: Music Map Clusters created with PCA projection on 2 dimensions

### 3 Predictive Modeling

As a **first model**, a hybrid content-based recommendation engine was developed by combining K-Means clustering (see Figure 6) with Cosine Similarity. The model relies on ten specific audio features: acousticness, danceability, energy, instrumentality, liveness, speechiness, tempo, popularity, loudness and valence. Which were selected because they represent the "musical DNA" of a track. Unlike metadata like popularity or release year, these variables capture the intrinsic mood, texture, and rhythmic intensity of the audio itself, allowing the model to focus on true sonic similarity. By first categorizing the dataset into fifteen distinct "musical families" using K-Means, the system narrows the search space to relevant neighbors, after which Cosine Similarity is applied to identify the most mathematically similar tracks within those groups (not considering songs from the same artist).

As a **second model**, a K-Nearest Neighbors (KNN) with Euclidean distance algorithm was implemented to evaluate tracks within a multidimensional space defined by the same pre-scaled audio features. Rather than clustering the dataset beforehand, this model maps each track as a distinct point in space to measure the absolute "physical proximity" between songs. By utilizing Euclidean distance as the core metric, the system prioritizes technical precision and high stability. This mathematical approach calculates the direct straight-line distance between feature vectors, allowing the engine to isolate tracks with nearly identical numerical profiles. Consequently, it ensures exceptional consistency in the technical audio characteristics of the recommendations, reliably grouping tracks that share an identical structural composition.

### 4 Evaluation and Comparison

To verify the accuracy and versatility of the recommendation engine, we conducted a qualitative evaluation using a diverse multi-genre test set (including tracks like *"Beat It"*, *"Yesterday"*, and *"Bohemian Rhapsody"*). The goal was to observe how the algorithm responds to distinctly different audio profiles.

A fascinating result emerged when testing the high-energy dance track *"On The Floor"* on the hybrid model. The system successfully recommended logical matches such as *"Up"* by Justin Bieber and *"Otra Noche"* by Proyecto Uno, sharing the exact same club-driven sonic footprint. However, it also surprisingly recommended Country and Blues-Rock tracks like *"Dream Walkin'"* by Toby Keith. While counter-intuitive to human cultural perception, this highlights the objective strength of our content-based model: the algorithm is completely blind to genre labels and social biases. It relies solely on the mathematics of sound, recognizing that a fast-paced, heavily strummed acoustic guitar can generate the same mathematical "Energy" and "Tempo" as an electronic bassline. This phenomenon allows the system to achieve high *serendipity*, recommending mathematically twin tracks across completely unrelated genres.

A direct comparison between the two models further highlights their structural differences, and becomes particularly tangible when examining the diverging outputs for Michael Jackson's *"Beat It"*. The hybrid model returned tracks such as *The One That Got Away* (Katy Perry), *What Is Love* (Haddaway), and *Our House* (Madness), all with cosine similarity scores exceeding 0.998, reflecting a tight directional alignment within the assigned cluster. The KNN model, operating globally via Euclidean distance, instead surfaced *All About That Bass* (Meghan Trainor), *Toxic* (Britney Spears), and *La Mordidita* (Ricky Martin), with distance scores between 0.125 and 0.142, tracks from entirely different genres, yet sharing a closely aligned normalized audio profile. This contrast effectively illustrates the core trade-off between the two approaches: the hybrid model prioritizes cluster-bound stylistic consistency, while the KNN model favors global mathematical precision, making it a valuable complementary tool for discovering unexpected but numerically grounded musical affinities.

**CRedit author statement:**

**Pietro Forni, Manuel Tagliaferri:** Conceptualization, Methodology, Software; **Pietro Forni:** Data curation, Writing-Original draft preparation, Supervision; **Pietro Forni, Manuel Tagliaferri:** Visualization, Investigation, Editing; **Manuel Tagliaferri:** Writing- Reviewing.